



19. Adaptogenic Plants

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Introduction

The term adaptogen was originally coined in 1947 by the pharmacologist N.V. Lazarev to describe the unexpected effect of dibazol (2-benzylbenzimidazol), an arterial dilator developed in France. Dibazol was found to increase the resistance of organisms to stress in experimental studies. Adaptogen derives from the Latin *adaptare* meaning to adjust or fit, and 'gen' from the Greek word *genes* meaning born of, or produced by. Incidentally, it is generally regarded as a barbarism to combine Greek and Latin roots in a single word.

The definition of adaptogens is as follow: (i) the adaptogenic effect is non-specific in that the adaptogen increases resistance to a very broad spectrum of harmful factors of different physical, chemical and biological natures; (ii) an adaptogen has a normalizing effect, that is, it counteracts or prevents disturbance brought about by stressor; (iii) an adaptogen must be innocuous to have a broad range of therapeutic effects without causing disturbances to the normal functioning of the organism.

Adaptogens, thus defined, constitute a new class of metabolic regulators (of natural origin) which have been shown to increase the ability of the organism to adapt to environmental factors and to avoid damage from such factors. Table 19.1 lists some of these plants. Adaptogen as a new concept has become more generally recognized during the last 10 years and has recently been used as a functional term by health authorities such as the FDA. Box 19.1 illustrates the difference between adaptogens and stimulants.

Table 19.1
Herbal medicines promoted as adaptogenic

Common name	Latin name	Part(s) of plant used	Key constituents	Daily dose
Ashwagandha	<i>Withania somnifera</i>	Roots	Alkaloids, steroidal lactones	3–6 g
Astragalus	<i>Astragalus spp.</i>	Roots	Flavonoids, polysaccharides	2–6 g
Cat's claw	<i>Uncaria tomentosa</i>	Root bark	Triterpenes, organic acids, glycosides, procyanidins, alkaloids	0.3–1 g
Cordyceps #	<i>Cordyceps sinensis</i>	Fruiting body#	Polysaccharides, nucleoside fractions	3–9 g

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Table 19.1

Herbal medicines promoted as adaptogenic (*continued*)

Common name	Latin name	Part(s) of plant used	Key constituents	Daily dose
Ginseng*	<i>Panax ginseng</i>	Roots	Triterpene saponins	1–2 g
Golden root	<i>Rhodiola rosea</i>	Roots	Organic acids, flavonoids, tannins, phenolic glycosides	a
Gotu kola	<i>Centella asiatica</i>	Leaves	Triterpenes, flavonoids, volatile oil	1.8 g
Sarsaparilla	<i>Smilax spp</i>	Roots	Steroid saponins	0.3–1.5 g
Schizandra	<i>Schizandra chinensis</i>	Fruits	Volatile oil, ascorbic acid, lignans	1.5–6 g
Siberian ginseng (eleuthero)*	<i>Eleutherococcus senticosus</i>	Roots, rhizome	Eleutherosides, carbohydrates, caffeic acid derivatives	2–3 g
White bryony	<i>Bryonia alba</i>	Roots	Terpenoids, steroids, fatty acids	0.3–0.5 g

* supported by German Commission E for lack of stamina

C. sinensis is a fungus

a. an extract standardised to contain 1% of the glycosides rosavin is generally used (100–170 mg/daily)

Table 19.2

True ginseng (*Panax ginseng*) and various succedaneous

Common name	Botanical name
Korean or Chinese or Asian ginseng	<i>Panax ginseng</i>
Japanese ginseng	<i>Panax japonicus</i>
American ginseng	<i>Panax quinquefolium</i>
San-chi or Tien-chan ginseng	<i>Panax notoginseng</i>
Vietnamese ginseng	<i>Panax vietnamensis</i>
Alaskan ginseng	<i>Echinopanax horridum</i>
Siberian ginseng (eleuthero)	<i>Eleutherococcus senticosus</i>
Indian ginseng (ashwagandha)	<i>Withania somnifera</i>
Brazilian ginseng (suma)	<i>Pfaffia paniculata</i>
Peruvian ginseng (maca)	<i>Lepidium meyenii</i>
Wild ginseng	<i>Aralia nudicaulis</i>
Red wild ginseng	<i>Rumex hymenosepalu</i>

Adaptogenic Plants

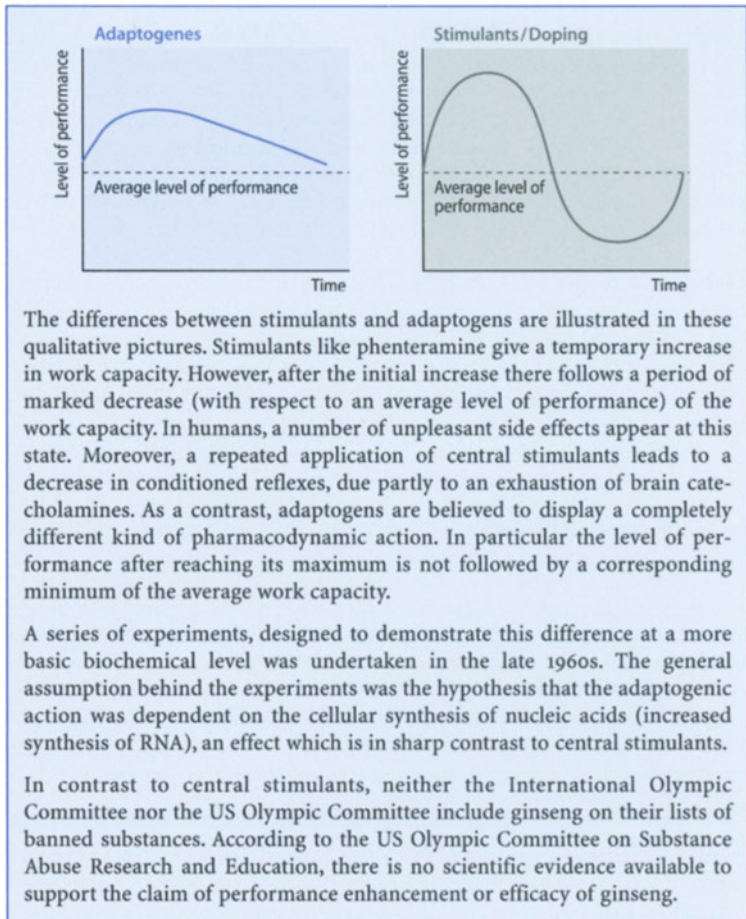


Ginseng

Botany/Key constituents ► Ginseng is the root of the perennial herb of a plant of the genus *Panax* (Fam. *Araliaceae*). Three medicinal species of ginseng are currently recognized: *P. ginseng* C.A: Meyer (Chinese or Korean ginseng, a perennial herb indigenous to the mountainous forests of Eastern Asia); *P. japonicus* C.A:

Box 19.1

Differences between stimulants and adaptogens



Meyer (Japanese ginseng from Vietnam, Southern China and Japan); and *P. quinquefolius* (American ginseng, found growing in rich woodland in the Eastern and Central USA and Canada). The most commonly used is *Panax ginseng* (true ginseng, see also Table 19.2), a perennial slow-growing herb (Plate 19.1). The main active constituents of the *Panax* species are recognized to be triterpenoid glycosides or saponins, termed ginsenosides, whose structures and distribution vary

with species and variety. There are twenty-five ginsenosides that have been separated and detected based on the sugar unit sequences and aglycone moieties.

Mode of action ► Experimental studies have confirmed the adaptogenic properties of ginseng in several stress models used to establish an adaptogenic effect. Ginseng may enhance resistance to X-irradiation, viral and tumor load, temperature stress, hyperbaric hypoxia, and physical exercise. Some studies report that ginseng augments work capacity and increases swimming time in rats. Many of these activities have been attributed to a corticosteroid-like action and endocrinological studies have suggested that ginsenosides may augment adrenal steroidogenesis via an indirect action on the adenohipophysis (Figure 19.1).

Clinical studies ► German Commission E monograph on ginseng states that the herb is used “as a tonic to counteract weakness and fatigue, as a restorative for declining stamina and impaired concentration, and as an aid to convalescence”. Ginseng is promoted to increase physical performance, psychomotor performance and cognitive function, immunomodulation, diabetes, and erectile dysfunctions. However, the evidence for ginseng’s efficacy is not compelling for any of these indications (Box 19.2).

Adverse events/Contraindications ► When used appropriately, ginseng appears to be relatively safe. Nevertheless, documented side effects include hypertension, diarrhea, insomnia, mastalgia, and vaginal bleeding. The long-term use of ginseng could cause central nervous system excitation and arousal (referred to as ginseng abuse syndrome). A dose of 15 g/daily has been associated with depersonalization, confusion and in some cases depression. There is also evidence that ginseng could cause Steven-Johnson syndrome. The German Commission E recommends that the use should be limited to 3 months. Documented herb/drug interactions include maniac symptoms, headaches, hallucinations and insomnia when ginseng is co-administered with the antidepressant phenelzine and over-anticoagulation when given together with the anticoagulant warfarin.

Preparations/Dosage

Ginseng can be administered as an infusion, powder and galenic preparation for internal uses. According to German Commission E the recommended daily dosage is 1–2 g crude drug. Many clinical studies have used a daily dose of 200 mg of a concentrated aqueous standardized extract (extract G115 standardized to 4% ginsenosides; each capsule is equivalent to 500 mg of *P. ginseng* root). It is supplied as capsules, liquid extract (1:2) and tablets.

- See Chapter 14 for the use of ginseng in the treatment of angina
- See Chapter 15 for the use of ginseng in the treatment of diabetes mellitus
- See Appendix B for the use of ginseng in cancer prevention

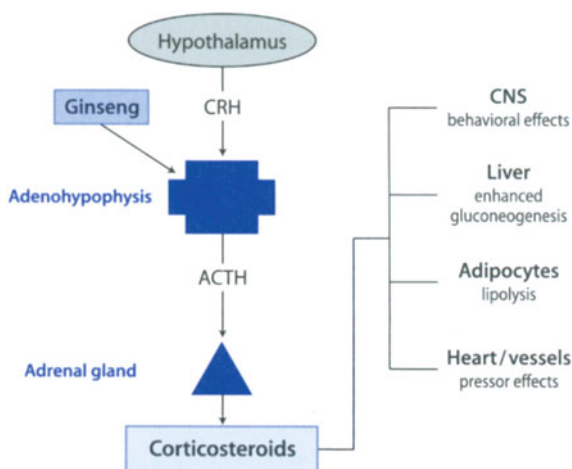


Figure 19.1 ▲ **Possible mechanism of ginseng in experimental studies**

Ginseng may augment adrenal synthesis of corticosteroids via an action on the hypothalamus/adrenohypophysis axis. Corticosteroids are known to play a significant role in the adaptation capabilities of the body to stress. Corticosteroids have several targets including the central nervous system, adipocytes, liver, heart and vasculature. CRH, corticotrophin-releasing hormone; ACTH, adrenocorticotrophic hormone; CNS, central nervous system

Eleuthero (Siberian ginseng)

Botany/Key constituents ► Eleuthero (also named Siberian ginseng) consists of the dried roots and/or rhizome of *Eleutherococcus senticosus* (Fam. *Araliaceae*). *Eleutherococcus* (from the Greek *eleutheros* meaning “free”, and *kokkos* meaning ‘pip’ or ‘seed’) *senticosus* is a thorny shrub that grows in the Russian Far East, the Amur Region, Primorsky Krai, Sakhalin, and also in North-east China, Korea and Japan. The thorniness, reflected by the specific epithet *senticosus*, an adjective meaning in Latin ‘full of briars or thorns’ has led to the common name in Russian of ‘thorny *Eleutherococcus*’ (*eleutherokokk koljuchii*). Key constituents include eleutherosides A-M (an heterogeneous group of compounds including sterols, phenylpropanoids, coumarins, monosaccharides and lignans), carbohydrates and caffeic acid derivatives (including chlorogenic acid).

Mode of action ► The adaptogenic properties of eleuthero has been extensively investigated in the ex-Soviet Union and pharmacological studies started in the 1950s. Nevertheless, eleuthero is reported to have immune-stimulating, antiviral, antioxidant, anticancer and hypocholesterolemic activities. In various stress models, the endurance of rodents was enhanced. In healthy volunteers, the lym-

Box 19.2

Ginseng and eleuthero: randomized clinical studies¹**Physical performance**

Seven trials investigated the effect of ginseng on physical performance in young, active volunteers during submaximal and maximal exercises on cycle ergometers. The four most recent studies found no improvement of physical performance after ingestion of *Panax ginseng*, *Panax quinquefolium* or *Eleutherococcus senticosus*. Other studies found a significant decrease in heart rate and an increase in maximal oxygen uptake compared with placebo, which persisted for three weeks after the treatment.

Psychomotor performance and cognitive function

Five studies investigated the effects of ginseng on psychological functions. Four of these were conducted on young, healthy volunteers while the only non-placebo-controlled trial was performed on elderly people. Three studies reported significant improvements in mental arithmetic and abstraction tests with *Panax ginseng* and in selective memory tests with *Eleutherococcus*. Four studies investigated whether ginseng may alter psychological functions and improve tolerability to exercise-induced stress. None of these studies confirmed this hypothesis. The results suggest no significant effects on the ratings of perceived exertion during cycle ergometer tests.

Immunomodulation

Two studies assessed the effects of ginseng on the immune system in healthy volunteers. One study reported a significant increase of the total number of T-lymphocytes and of the activity of leukocytes compared with baseline after the ingestion of *Panax ginseng*. A more recent study, however, found no effects on total and differential leukocyte counts and lymphocyte subpopulations.

Diabetes

One study investigated the effect of ginseng on Type-II diabetes mellitus. Patients received either 100 mg or 200 mg ginseng daily. At the end of an 8-week treatment period, psychophysical performance, mood and vigor were significantly improved compared with baseline in both ginseng groups. HbA_{1c} was significantly reduced in patients who received 200 mg ginseng extract, while a reduction of the fasting blood glucose level was observed in both ginseng groups compared with baseline.

Herpes infection

One study investigated the effect of *Eleutherococcus senticosus* on herpes simplex Type-II infection. Administration of 400 mg standardized *Eleutherococcus* extract once daily for 6 months produced a beneficial effect on the severity and duration of infections.

¹ From Vogler et al. 1999

phocyte count, especially that of T-lymphocytes, increased after eleuthero administration. Hypoglycemic effects have been also demonstrated in the herb along with enhancement of a platelet aggregation-inhibiting effect. The exact mechanisms responsible are not known.

Clinical efficacy ► Eleuthero is traditionally used as a tonic for invigoration and fortification in times of fatigue and debility or declining capacity for work and concentration, and during convalescence. It has been used as an inexpensive substitute for the ginseng roots of *Panax ginseng* in the republic of the former Soviet Union and other countries. The German Commission E monograph on eleuthero recommends its use “as a tonic to counteract weakness and fatigue, as a restorative for declining stamina and impaired concentration, and as an aid to convalescence”. However, clinical evidence for randomized clinical studies is scarce (Box 19.2).

Adverse events/Contraindications ► Health risks or side effects following the proper administration of designated therapeutic dosages are not recorded. As with ginseng, the German Commission E recommends that the use should be limited to 3 months. It is stated that eleuthero should be avoided by individuals who are highly energetic, nervous, tense, hysteric, manic or schizophrenia, and that it should not be taken with stimulants, including coffee, antipsychotic drugs or during treatment with hormones. In view of documented pharmacological actions eleuthero may theoretically interfere with a number of therapies including, cardiac, anihypertensive, anticoagulant and hypoglycemic therapy. The German Commission E notes hypertension as a contraindication. Eleuthero may interfere with plasma digoxin assay.

Preparations/Dosage

The recommended daily dose is 2–3 g crude root or an equivalent dose of an extract-based preparation. Extracts are normally standardized to contain more than 1% eleutheroside E. The extract is prepared from root powder and 75% ethanol (1:7) with back flow to which is added a 10% alpha-naphthol solution until there is no reaction. It is then evaporated to a paste.



Ashwagandha

Ashwagandha is the root of *Withania somnifera* (Fam. *Solanaceae*) and is sometimes erroneously referred to as Indian ginseng because of its popular use similar to that of ginseng. *Withania somnifera* is a small shrub widely distributed throughout the drier regions of India. The chemistry of Ashwagandha has been extensively studied and over 35 chemical constituents have been identified. The biologically active chemical constituents are alkaloids

(isopelletierine, anafherine), steroidal lactones (withanolides, withaferins) and terpenoids with a tetracyclic skeleton like cortisol (sitoinosides I-IV). Ashwagandha is also rich in iron.

Mode of action ► Many pharmacological studies have been conducted to investigate the properties of ashwagandha in an attempt to authenticate its use as a multi-purpose medical agent. Experimental studies have shown that ashwagandha possesses anti-inflammatory, antitumour, antioxidant, hemopoietic and immunomodulant properties. It also appears to exert a positive influence on the endocrine, cardiopulmonary, and central nervous systems. Moreover, ashwagandha induces a state of non-specific increased resistance during stress, sitoinosides being the active ingredients. The mechanisms of action for these properties are not fully understood.

Clinical efficacy ► Ashwagandha is widely used in Ayurvedic medicine, the traditional medical system of India. It is an ingredient in many formulations prescribed for a variety of musculoskeletal conditions (e.g. arthritis, rheumatism), and as a general tonic to increase energy, improve overall health and longevity, and prevent disease in athletes, the elderly and during pregnancy. However, there is a lack of randomized clinical studies to support the adaptogenic use. One double-blind clinical trial in 141 healthy male volunteers (aged 50–59), showed that ashwagandha, given for a period of one year, improved the hematological profile, lowered serum cholesterol and had other positive effects.

Adverse events/Contraindications ► No health hazards or side effects are known in conjunction with the proper administration of designated therapeutic dosages. However, high doses of the root in mice (750 mg/kg daily for 15 days) caused serious side effects, including diarrhea, weight loss, and death.

Preparations/Dosage

The usual daily dose of the dried powdered root is 3–6 g. It may be consumed in the form of decoction (2 g in 150 ml water for 5 min, about half an hour before meal time), liquid extract or tablets. A liquid extract (1:2, 6–12 ml) preparation is also available.

Other Adaptogenic Plants

Schizandra (Wu-Wei-Zi). Schizandra fruit is the dried, ripe fruit of *Schizandra chinensis* (Turez.) Baillon (Fam. *Schizandraceae*), a liana plant indigenous to North-Eastern China and Korea. The fruit is harvested in Autumn, then steamed before being dried in the sun. Key constituents include a volatile oil, vitamin C and lignans (schizandrine A to C, schizandrol A and B). Schizandra fruit and seeds are believed to bring about a non-specific increase in physical performance ability. Schizandra fruit activate formation of both nitric oxide and cortisol in blood

plasma and saliva and this activation is believed to “strengthen” the organism. Lignans are believed to be active principles with adaptogenic properties. Schizandrine and schizandrol possess liver-protective effect, have anti-inflammatory and antitumor activity in animal studies. Despite claims of efficacy, there are no clinical studies to support the adaptogenic properties of schizandra. No specific side effects are known in conjunction with the proper administration of schizandra fruit. Schizandra fruit is generally administered as whole and powdered drug for internal use. The daily dose is 1.5–6 g.

► See Chapter 22 for the hepato-protecting properties of schizandra

White bryony. Bryony consists of the dried tap-root of *Bryonia alba* L. (Fam. *Cucurbitaceae*), a plant indigenous from North-Eastern and South-Eastern Europe and also Iran (Plate 19.2). *Bryonia alba* is an extremely fast-growing perennial plant. It has a thick, tuberous root. The root is fleshy, wrinkled horizontally, yellowish-grey on the outside and white and slimy on the inside. Key constituents include terpenoids with a tetracyclic skeleton like cortisol (cucurbitacins), steroids and polyhydroxy fatty acids. White bryony has been used prophylactically and therapeutically for metabolic disorders, liver diseases, acute and chronic infectious diseases and to increase stress resistance. Like lignans of schizandra, cucurbitacins activate formation of both nitric oxide and cortisol and this activation is believed to adapt the organism to physical stress. Moreover, white bryony root has been used as a strong purgative and emetic. The drug is highly toxic when freshly harvested. The toxicity of the drug declines rapidly with dehydration and storage because of the instability of the cucurbitacins. Due to the cucurbitacin content, the drug has a severely irritating effect on skin and mucous membranes. The daily dosage is 0.3–0.5 g. The intake of high dosage can lead to vomiting, bloody diarrhea, colic, kidney complaints, paralysis and, under certain conditions, to death.

Cat's claw is the root bark of *Uncaria tomentosa* L. (Fam. *Rubiaceae*), a large woody vine that sometimes reaches heights of 30 m, indigenous to the rain forest areas of Central and South America. A less common *U. guaianensis* is also used. Key constituents include alkaloids (rynchophyllin, 5-carboxystrictosidine, isopteropodine), triterpenes, organic acids, sterols and procyanidins. Cat's claw possesses several pharmacological actions including anti-inflammatory/immunostimulating effects, inhibitory effect on platelet aggregation, contraceptive effects, antihypertensive and anticancer effects. Moreover, rynchophylline increases the level of neurotransmitters such as serotonin and dopamine in several brain areas. Cat's claw has been used in folk medicine for rheumatic complaints, diarrhea, gastritis, treatment of wounds, as an adjunct to cancer treatment, menstrual irregularity and as a contraceptive. A case report of a cancer patient in Austria who underwent a miraculous recovery helped bring attention to cat's claw in the 1970s. There are European reports that it is useful in the treatment of AIDS when

used in combination with zidovudine. However, cat's claw is probably now best classified as an adaptogen, although randomized clinical trials are lacking. Despite its extreme popularity in Hispanic communities, cat's claw cannot presently be viewed as a safe and effective phytomedicine. Serum estradiol and progesterone levels may be reduced after long-term cat's claw use and its use has been associated to acute renal failure. The daily dosage is up to 1000 mg (the total alkaloid equivalent should be 10–30 mg). Cat's claw is available in 300 mg capsules to be taken three times daily, liquid concentrate (8:1 in 20% alcohol) to be diluted in water and taken 1–3 times daily, and as a crude drug to be used for tea.

Golden root. The root of *Rhodiola rosea* (golden root or arctic root), is a plant widely distributed at high altitudes in Arctic and mountainous regions throughout Europe and Asia. It contains a range of biologically active substances including organic acids, flavonoids, tannins, and phenolic glycosides. It is a popular herbal drug in traditional medical systems in Eastern Europe and Asia, with a reputation for stimulating the nervous system, decreasing depression, enhancing work performance, eliminating fatigue, and preventing high altitude sickness. Golden root has been categorized as an adaptogen by Russian researchers due to its observed ability to increase resistance to a variety of chemical, biological, and physical stressors. The adaptogenic properties of golden roots have been attributed to p-tyrosol and the phenolic glycoside rhodioloside, which influence levels and activity of monoamines and opioid peptides in the brain and in peripheral tissues such as the cardiovascular and respiratory systems. Golden root is stated to be safe; however, some individuals might experience an increase in irritability and insomnia within several days. An extract standardized to contain 1% of the glycoside rosavin is generally used (100–170 mg/daily).

Further Reading

- Bahrke MS, Morgan WP (2000) Evaluation of the ergogenic properties of Ginseng. *Sports Med* 29:113–133
- Bucci LR (2000) Selected herbals and human exercise performance. *Am J Clin Nutr* 72:624S–636S
- Davydov M, Krikorian AD (2000) *Eleutherococcus senticosus* (Rupr. & Maxim.) Maxim. (*Araliaceae*) as a adaptogen a closer look. *J Ethnopharmacol* 72:345–393
- Kelly GS (2001) *Rhodiola rosea*: a possible plant adaptogen. *Altern Med Rev* 6:293–302
- Kitts D, Hu C (2000) Efficacy and safety of ginseng. *Public health Nutr* 3:473–485
- Mishra LC, Singh BB, Dagenais S (2000) Scientific basis for therapeutic use of *Withania somnifera* (ashwagandha): a review. *Altern Med Rev* 5:334–346
- Murphy LL, Lee TJ (2002) Ginseng, sex behavior, and nitric oxide. *Ann N Y Acad Sci* 962:372–377
- Nocerino E, Amato M, Izzo AA (2000) The aphrodisiac and adaptogenic properties of ginseng. *Fitoterapia* 71:1–5
- Panossian A, Wikman G, Wagner H (1999) Plant adaptogens III. Earlier and more recent aspects and concepts on their mode of action. *Phytomedicine* 6:287–300
- Rege NN, Thatte UM, Dahanukar SA (1999) Adaptogenic properties of six rasayana herbs used in Ayurvedic medicine. *Phytother Res* 13:275–291
- Thomson Coon J, Ernst E (2002) *Panax ginseng*: a systematic review of adverse effects and drug interactions. *Drug Saf* 25:323–344
- Vogler BK, Pittler MH, Ernst E (1999) The efficacy of ginseng. A systematic review of randomized clinical trials. *Eur J Clin Pharmacol* 55:567–575